

Wireless Communication Exam 1

Note**

1. Each question carries 10 marks
2. Follow the format for submission

Student Name: jkl

Email: jkl@gmail.com

Student ID: 321

1. Define modulation and explain its purpose in wireless communication.

Answer: Modulation is the process of increasing the power of a signal to ensure it can travel longer distances without attenuation. Its primary purpose in wireless communication is to amplify signals for better reception.

Score: 1.5/10

Justification: The student's answer lacks clarity, organization, and understanding of modulation techniques in wireless communication. The answer fails to provide examples, compare and contrast different modulation techniques, and lacks accuracy.

Feedback: The student should provide clear and concise answers, well-organized and easy to understand. They should provide examples of different modulation techniques, their advantages and disadvantages, and how modulation is used in wireless communication systems. The student should also provide a clear and accurate definition of modulation and explain its purpose in wireless communication, as well as provide examples of how modulation is used to encode and transmit information over different frequencies.

2. Calculate the path loss at a distance of 500 meters for a signal frequency of 2 GHz in free space.

Answer: The path loss at 500 meters for a 2 GHz signal in free space is approximately 50 dB .

Score: 1.5/10

Justification: The student's answer lacks understanding of the concept of path loss and its calculation, with incorrect calculations and no explanation of its significance in wireless communication systems.

Feedback: The student should review the concept of path loss and its calculation to better understand the topic, and carefully follow the formula to get the correct answer.

Summary:

Total Score: 3.0

Percentage: 15.00%

Grade: F

Overall Feedback: To improve, the student should focus on providing clear and concise answers that are well-organized and easy to understand. They should provide specific examples of modulation techniques, their advantages and disadvantages, and explain how modulation is used in wireless communication systems. Additionally, the student should review the concept of path loss and its calculation to ensure accurate answers, and carefully follow the formula to get the correct result.